

Kevin Xu

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Professional Summary

Data Science Consultant and Statistics M.A. graduate from UC Berkeley (May 2026) with 2+ years of experience building Python, PySpark, and SQL data pipelines and statistical models. Experienced in anomaly detection, regression, time-series forecasting, feature attribution, and production-oriented data validation across large financial datasets. Built an anomaly detection engine that surfaced ~\$22M in recoverable assets and brings additional experience in interpretable ML and LLM systems.

Education

University of California, Berkeley

Master of Arts, Statistics | **Achievements:** Elizabeth Scott Excellence Scholarship

Sep 2024 - May 2026

University of California, Davis

Bachelor of Science, Statistics | **Achievements:** Dean's List, 6 terms

Sep 2020 - Jun 2022

Work Experience

Deloitte, San Francisco

Consultant (Data Science)

Aug 2022 - Apr 2025

- Designed and owned a rule-based anomaly detection engine across a \$12.2B defense financial portfolio, automating pattern recognition for valuation mismatches, duplicate records, and cross-system breaks while surfacing ~\$22M in recoverable assets.
- Re-engineered 8+ legacy reconciliation workflows into modular PySpark and Databricks data pipelines with explicit schema contracts and execution gates, reducing processing latency from hours to minutes and ~80% runtime reduction, under strict audit constraints.
- Architected batch ingestion pipelines that consolidated multi-source data streams into a unified AWS S3 Data Lake; built pre-warehouse validation layers, schema enforcement, and threshold-based quality gates to prevent corrupted data from propagating downstream.
- Engineered a full-graph data lineage framework tagging records with transformation histories and validation outcomes, improving data pipeline transparency and supporting external audit compliance.
- Evaluated product roadmaps, prioritizing features, and aligned engineering spend with strategic goals while contributing to and leading R&D assessments.
- Authored system-level technical documentation and built real-time portfolio health dashboards to support business intelligence and data visualization.
- Mentored 4 junior data scientists on production workflows and pipeline reliability.

Data Science & AI Engineering Projects

Natural Language to fMRI Brain Signal Analysis

- Built an end-to-end Python pipeline to predict brain activity from story text, comparing six text-representation methods including Word-to-Vec, GloVe, and fine-tuned BERT against fMRI scans from two subjects with ~95,000 brain locations each.
- Found that only large pretrained language models carried meaningful signal: pretrained BERT increased mean correlation by ~60-90% over simpler models, while parameter-efficient-fine-tuning with LoRA rank-8 adapters added little. Used SHAP/LIME for model interpretation, identifying words associated with each brain region's response.

Natural Language Text-to-SQL Agent for Analytics

- Built an AI Agent that turns plain-English business questions such as "What was churn last month", into SQL.
- Designed a hallucination-resistant workflow that retrieves relevant schema and metric definitions through vector search, generates SQL, and self-corrects when validation fails rather than querying non-existent columns.
- Added column-level validation via SQL AST parsing (sqlglot), restricted queries to approved semantic views, and wrote adversarial tests blocking hallucinated columns and destructive operations.

Polar Cloud Detection via Autoencoders & Ensemble Classifiers

- Built a model to detect clouds in NASA polar satellite imagery (MISR multi-angle radiances) where clouds and snow/ice look nearly identical to tackle a known problem in remote sensing.
- Engineered spatial-neighborhood features and pre-trained encoders on 132 unlabeled images to learn compact latent representations, using image level splits to prevent leakage.
- Benchmarked RF (98.15%), KNN (94.45%), and LightGBM (98.74%) under quadrant-based spatial cross-validation. Noise injection stress-tests showed robustness to feature noise (98%) but collapse under raw-data noise (43%).

Skills

LLM & Agentic AI: LangChain, RAG Systems, Vector Databases (ChromaDB), Parameter-Efficient Fine-Tuning (PEFT/LoRA), SQL AST Parsing & Validation (sqlglot), Human-in-the-Loop Evaluation, Prompt Engineering.

Data & Cloud Engineering: PySpark, Databricks, AWS (Glue, S3, Athena, Step Functions), Data Modeling (dbt), Distributed Systems, ETL/ELT Pipelines, Pre-Warehouse Validation, Schema Enforcement and Deduplication, Data Pipelines, Relational Databases.

Languages & ML Frameworks: Python, SQL (MySQL, T-SQL), PyTorch, Hugging Face, Scikit-Learn, LightGBM, Pandas, NumPy, Statsmodels.

Statistical & ML Methods: Time-Series Forecasting (SARIMA), Feature Attribution (SHAP/LIME), Bayesian Optimization, Anomaly Detection, Spatial Cross-Validation, Regression Analysis, Machine Learning.

Developer Tools & Workflow: Docker, FastAPI, RESTful APIs & Webhooks, Git, Bash, Pipeline Profiling & Automated Alerting, Jupyter

Business Intelligence & Data Visualization: PowerBI, Python (ggplot2, matplotlib, seaborn, Tableau).